

Week 2 — Requirements Engineering

CSE5IDP Industry Development Project · Presenter: Scott Mann · Online cohort notes

Semester 2, 2026 · Milestone 1

Housekeeping and admin

- LMS updates: a link to the Semester 1 trade show (for inspiration) plus setup instructions for Semester 2 are now on the LMS homepage.
- Echo 360 recordings: if you cannot see the panel top-right, expand the small red tab, or go to Live Sessions, then Recordings. On a Mac, use Safari — it is the only browser he has found reliable for Echo 360.
- Jira invite emails are not arriving for some students; he has escalated it to the Atlassian administrator. Stick with it.
- Teams and Jira boards have been provisioned only for teams that have fully registered and voted on a project. No project selected means no board yet.
- Project owners are being added to the Teams channels from Thursday, so keep the language professional.
- Online students without a team should contact Ali (face-to-face students were told to see him at the break).

Trade show (week 4, Thursday 6 August)

- Two windows: 11:00–13:00 and 19:00–21:00. You do not need the whole team present at once — split into a morning shift and an evening shift.
- One person per team must email him to be nominated as the booth editor. Setup is about 15 minutes.
- Content should come out of Sprint 1: requirements reflected back, visual prototypes, Figma wireframes, short video summaries. He explicitly warned against “daggy” fully-AI-generated filler.

Sprint 1 — what you should be doing right now

You are technically already in Sprint 1. He named three parallel workstreams:

1. Build out the Jira product backlog — break the project description into functional requirements, create tasks, allocate them, and estimate and size them.
2. Prepare questions for the project owner meet-and-greet next week (schedule to be released).
3. Start producing trade show material.

He will be checking backlogs team-by-team in the labs, and will run a live Jira walkthrough — composing a backlog, sprint backlog, task allocation, sizing, estimation — using one real project as the worked example.

On stand-ups: 10–15 minutes, each person covers what they did, what they are doing, and what is blocking them. His point was that flagging a blocker is responsible, not weak — it lets the team absorb the work and keeps the burndown chart from flatlining.

Requirements engineering (the core content)

User stories

Three-part form: “as a *user* I want *action* so that *value*”. He recommended looking up the INVEST framework for structuring them properly. Stories decompose into tasks; stories that span multiple sprints

should be marked up as an epic. He suggested the physical card approach — UI sketch on the front, positive and negative acceptance criteria on the back — and said you are welcome to attach these as evidence on the Jira task.

Acceptance criteria must cover both normal and abnormal execution flows, with preconditions and postconditions. The technique he wants most is taking a user story all the way through to a test case — he will do this one-on-one with each team in the labs if you bring a functional requirement with you.

User flows

A user entering the system and completing a set of related tasks to get a job done. His worked example was a bank transfer: land on the app, authenticate, select source account, set up recipient, set amount, execute, get confirmation. Draw these out on A3 with the whole team — happy path, navigation points, validations, error states.

User versus system requirements

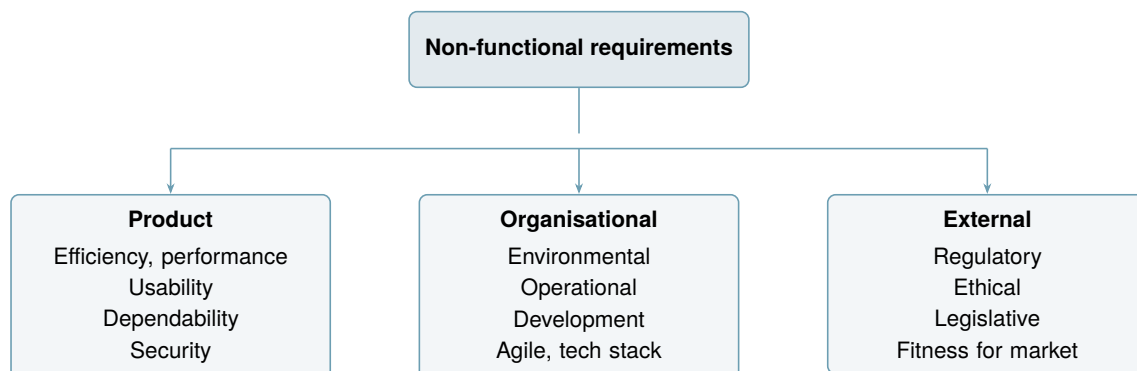
From Sommerville's medical patient management example:

- **User requirement** — natural language, outcomes-focused, written for customers and end users.
- **System requirement** — the technical elaboration: rules, timings, interfaces, third-party APIs, data formats, payload sizes.

His warning here: project owners vary enormously in technical depth. Some will have thought at the system level, some will expect you, as near-graduates, to define it for them.

Functional versus non-functional

This was the emphasis of the whole lecture. He called the non-functional taxonomy “one of the most important diagrams in this whole subject.”



His framing: a failure in any one of those three branches is what he called “the kiss of death” — the system may tick every functional box and still be unusable. The mandate for you: use IDP as an organisational constraint (agile scrum is imposed on you; your project owner may impose a tech stack or cloud platform on top), and remember that ethics and legislation sit outside your control but determine whether the product can be sold at all.

Two examples he used to make it concrete: data storage cost in 2026 is now a serious efficiency constraint (bloated, lossy storage is “shooting yourself in the foot”), and architecture churn — PowerPC to Intel to Apple Silicon — kills software that is not maintained forward.

Quantify your non-functional requirements

This was the “how to impress me” section. Do not just state NFRs — measure them. Speed, size, usability, reliability, robustness, portability. His worked scenario: prototype with a candidate stack in Sprint 1, then in week 4 or 5 swap out the database backend, measure a 20 per cent improvement, and report the delta. That becomes a story you can tell, and it is the kind of quantified evidence he says a modern portfolio needs — positions, plus responsibilities, plus hard data.

He suggested putting a modelled-and-validated NFR slide into the final presentation deck.

AI, and repositories

- Detailed, formal requirements are the foundation for agentic AI development. Vague prompts produce vague systems; graphical control flows and mathematical specs produce good ones. His line: it is still human-first — you teach the AI, you do not infer the specification from it.
- He dislikes the term “prompt engineer” — a useful skill, not a discipline or a certification.
- Handover quality matters to your grade. The project owner must be able to clone your repo and get it running with minimal effort: well-structured code, in-code documentation, and a real root [README](#). “It worked on my machine” will not fly.

Employability segment

- Download the ACS Digital Pulse report (2025 edition, free from the ACS site; 2026 edition due in October). He wants everyone to read it this semester and use it to target the job search.
- The three hot domains, repeated throughout: AI, cybersecurity, and data science / analytics and visualisation. Be a generalist by degree, a specialist by certification.
- Specific recommendation: the Microsoft Power Platform certification, not just Power BI.
- SFIA (Skills Framework for the Information Age) will be covered in the second half of IDP; you will build a SFIA-aligned CV.
- Also check the Victorian ICT / digital skills snapshot (other states publish equivalents) — “follow the money” to find where government funding is going.
- Certifications are baked into the assessment: he said it is literally the first rubric row of the scrum master presentation.
- Reference text: Sommerville, *Software Engineering* (11th or 12th edition) — the source of much of this week’s material.
- Honest framing: the current job market is the most chaotic he has seen in 16 years, expect 6–18 months, and note that Melbourne and Sydney are the only real tech hubs (Microsoft in Melbourne is sales, not development).

Notes specific to the online cohort

- The Thursday labs are face-to-face on campus and not normally streamed. He agreed to record this week’s as a one-off, but going forward Ali runs the equivalent two-hour session for you online on Wednesday nights.
- Post questions in the chat at any time, or unmute — he repeatedly invited online questions and answered several live.
- No team? Contact Ali.
- He flagged twice that online students should pick up the Teams-threading technique and the certification advice, since the on-campus students got them visually.

Action items

Item	When
Log into Jira, populate the product backlog, allocate and size Sprint 1 tasks	Before your lab or Ali’s session
Bring one functional requirement, decomposed into user requirements plus 3–4 system requirements	To the lab session
Nominate one team member as trade show booth editor and email him	As soon as possible
Prepare a dot-point question list for the project owner meet-and-greet	Before next week
Start Teams threads per topic; use @name to reach your project owner or @Scott for him	Now
Begin trade show booth material	Runs through Sprint 1

Download the ACS Digital Pulse report and start a certification plan

This semester, sooner is better

The single biggest emphasis of the session: non-functional requirements determine project success or failure, and quantifying them is what separates a passing team from a memorable one.